Strategy Report

Project: Precious Metal Premium Mean-Reversion Strategy

Last Update Time: 2026-03-18 15:43:34

Update Status: failed

Update Message: main.py □□□□□□□□ /Users/patric/Desktop/□□□□□/Precious_Metal_ETL/data/processed/daily_update_log.txt

Best Strategy Summary

- Asset: silver
- Signal: BUY
- Holding Days: 20
- Final Capital: 41702.5
- Total Return %: 317.02
- Trades: 18

Core Findings

- BUY strategies outperform SELL strategies in the last 2 years.
- Silver BUY strategies show the strongest returns.
- Silver BUY 20D is currently the best-performing strategy.

Backtest Summary Table

asset	holding_days	signal	avg_return_pct	win_rate_pct	trades	max_return_pct	min_return_pct
gold	5	BUY	0.82	62.5	16	8.46	-8.59
gold	5	SELL	1.03	66.67	3	2.73	-1.31
gold	10	BUY	1.18	62.5	16	13.09	-7.42
gold	10	SELL	-0.81	33.33	3	0.81	-3.13
gold	20	BUY	3.69	68.75	16	22.58	-2.97
gold	20	SELL	-2.41	66.67	3	2.06	-10.44
silver	5	BUY	2.62	55.56	18	15.28	-3.68
silver	5	SELL	-0.95	44.44	9	4.72	-6.15
silver	10	BUY	4.48	72.22	18	22.93	-4.51
silver	10	SELL	-2.77	33.33	9	7.23	-8.71
silver	20	BUY	9.1	83.33	18	61.92	-9.57
silver	20	SELL	-5.56	33.33	9	5.55	-23.76

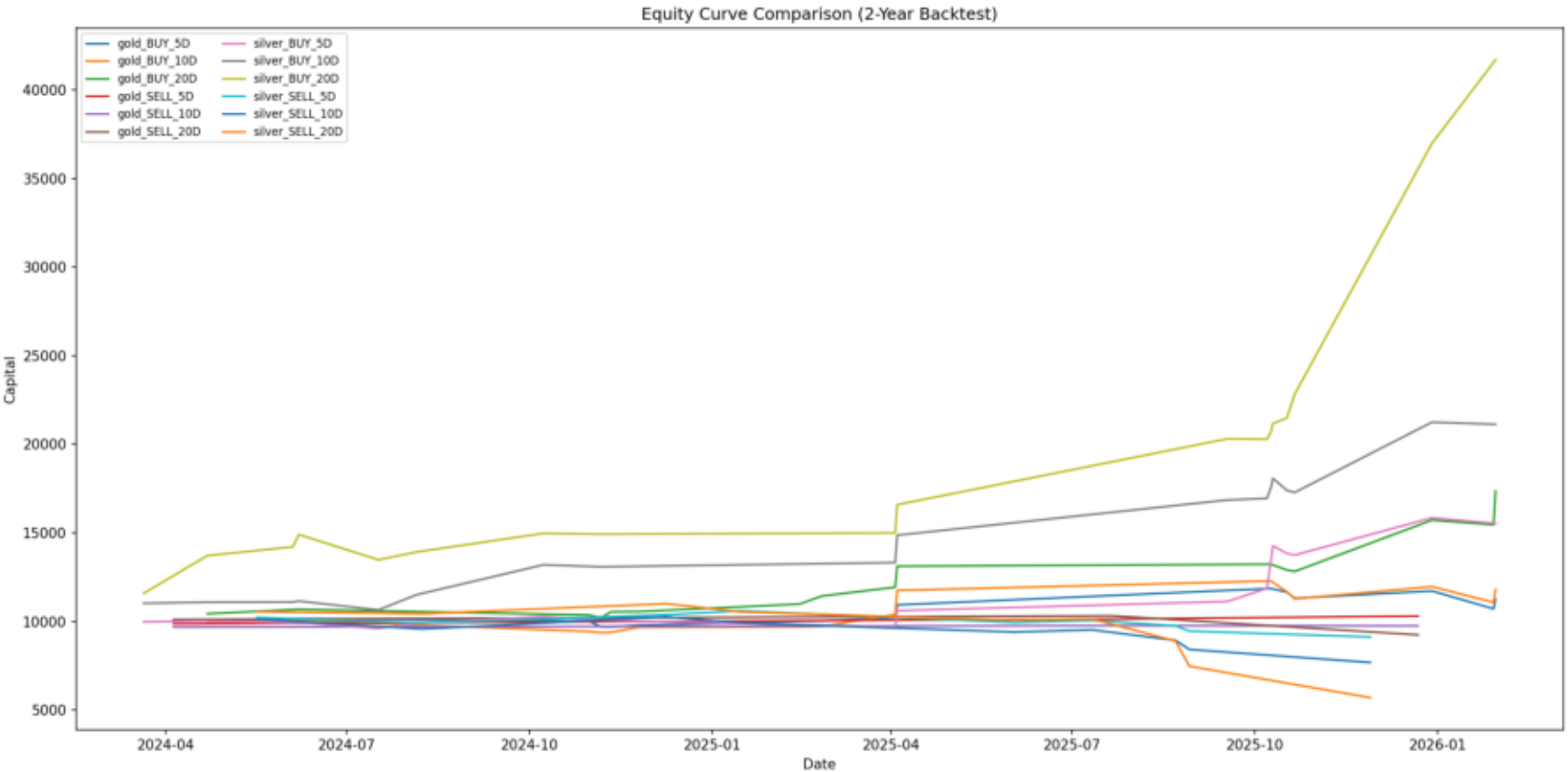
Risk Metrics Table

asset	signal	holding_days	trades	final_capital	total_return_pct	win_rate_pct	max_drawdown_p	volatility_pct	sharpe_ratio
silver	BUY	20	18	41702.5	317.02	83.33	-9.57	15.26	0.5967
silver	BUY	10	18	21129.45	111.29	72.22	-4.51	7.38	0.607
gold	BUY	20	16	17364.68	73.65	68.75	-4.46	6.59	0.5609
silver	BUY	5	18	15535.94	55.36	55.56	-5.63	5.67	0.4625
gold	BUY	10	16	11802.7	18.03	62.5	-9.84	5.4	0.2175
gold	BUY	5	16	11255.04	12.55	62.5	-9.66	4.14	0.1984
gold	SELL	5	3	10308.0	3.08	66.67	0.0	2.1	0.4919
gold	SELL	10	3	9755.39	-2.45	33.33	-0.11	2.06	-0.3923
gold	SELL	20	3	9245.33	-7.55	66.67	-10.44	6.97	-0.3458
silver	SELL	5	9	9126.87	-8.73	44.44	-13.89	3.59	-0.2652
silver	SELL	10	9	7687.32	-23.13	33.33	-25.13	5.04	-0.5491
silver	SELL	20	9	5701.32	-42.99	33.33	-48.21	9.94	-0.5595

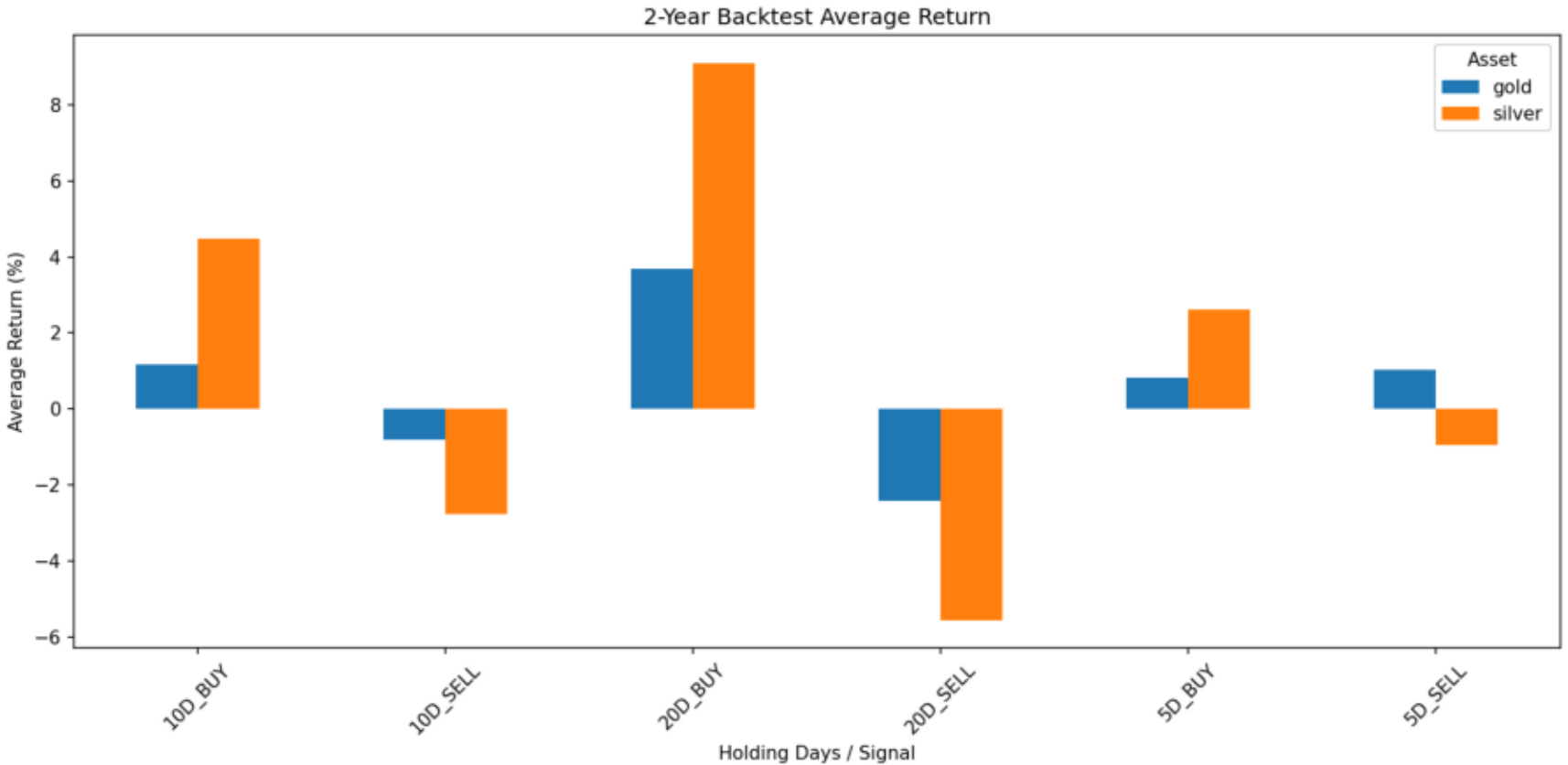
Strategy Ranking Table

asset	signal	holding_days	final_capital	trades	total_return_pct
silver	BUY	20	41702.5	18	317.02
silver	BUY	10	21129.45	18	111.29
gold	BUY	20	17364.68	16	73.65
silver	BUY	5	15535.94	18	55.36
gold	BUY	10	11802.7	16	18.03
gold	BUY	5	11255.04	16	12.55
gold	SELL	5	10308.0	3	3.08
gold	SELL	10	9755.39	3	-2.45
gold	SELL	20	9245.33	3	-7.55
silver	SELL	5	9126.87	9	-8.73
silver	SELL	10	7687.32	9	-23.13
silver	SELL	20	5701.32	9	-42.99

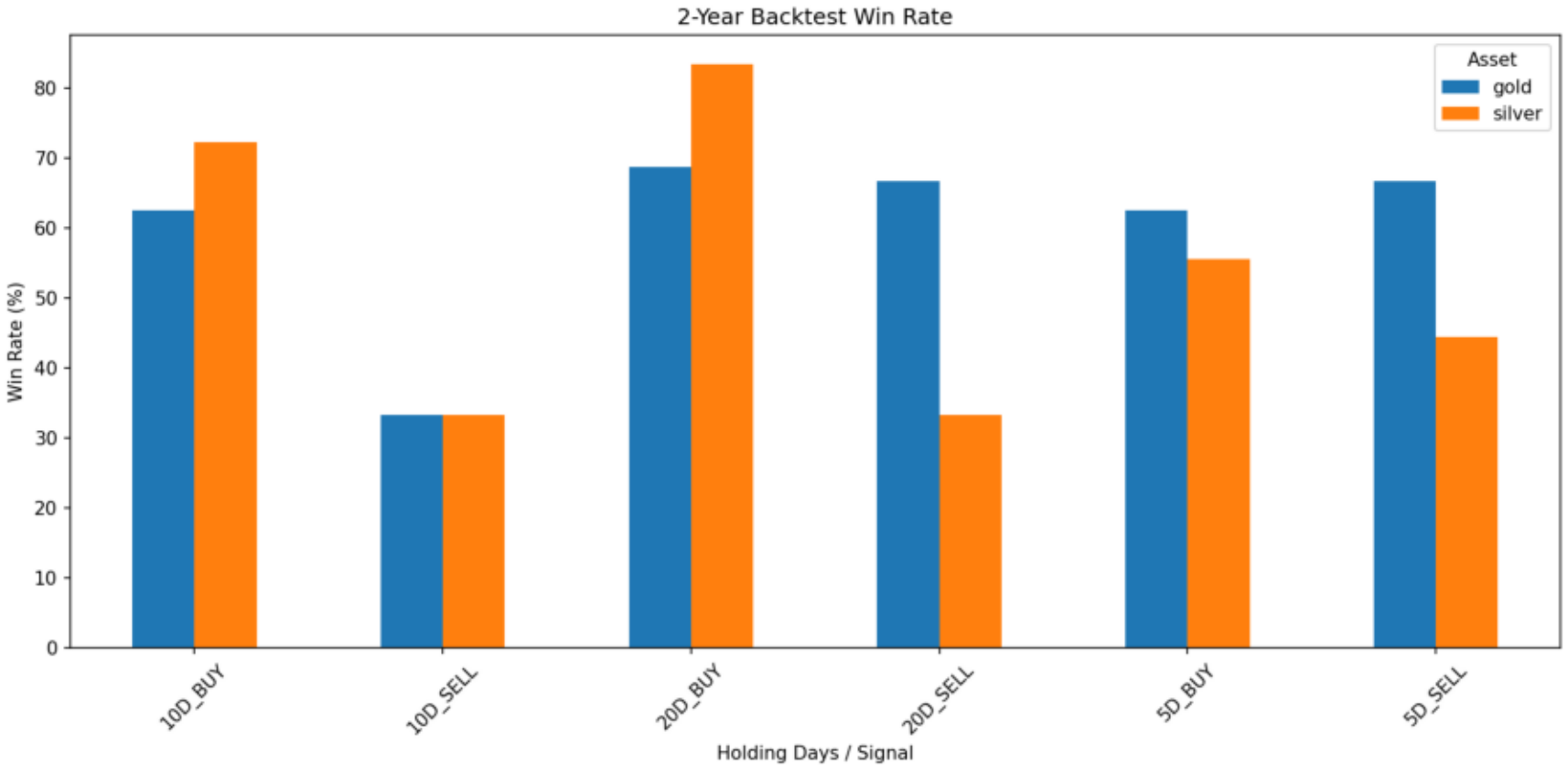
Equity Curve Comparison



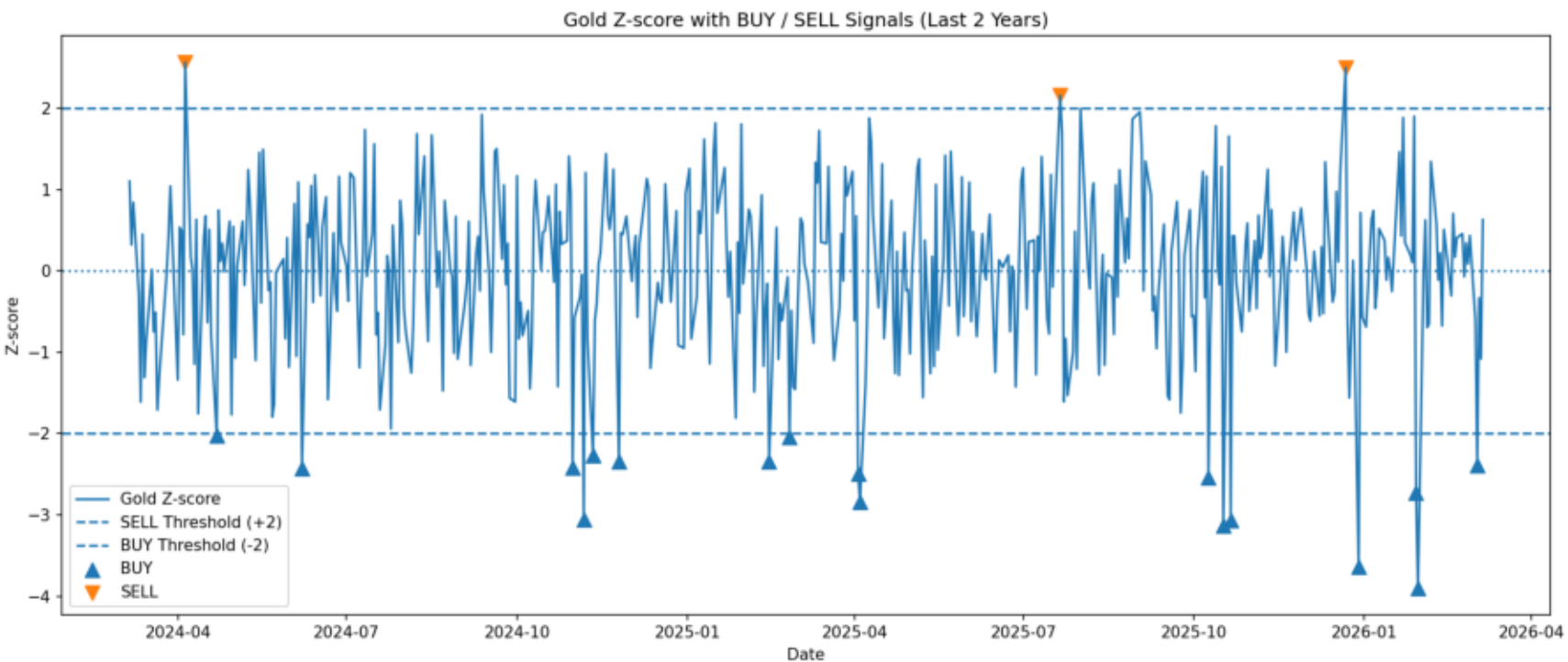
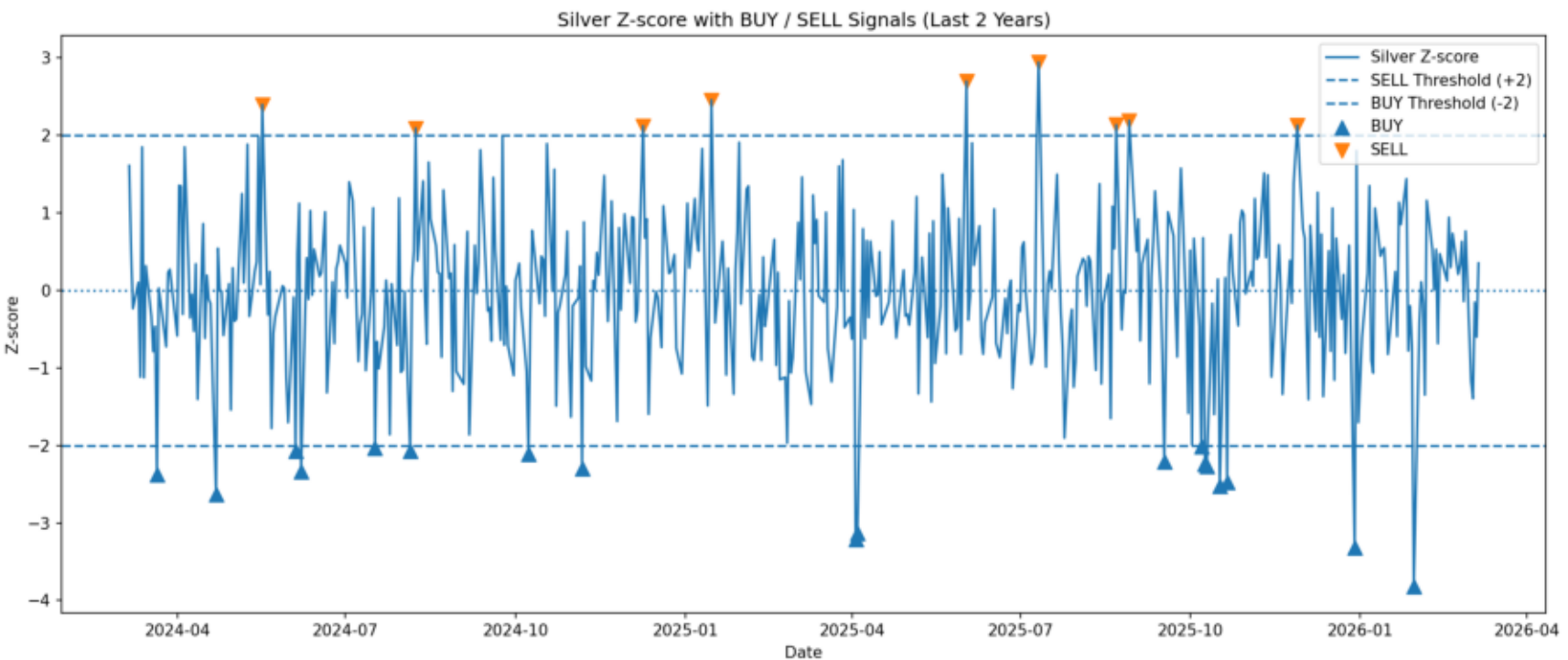
Backtest Average Return



Backtest Win Rate



Gold Z-score Signals



Silver Z-score Signals